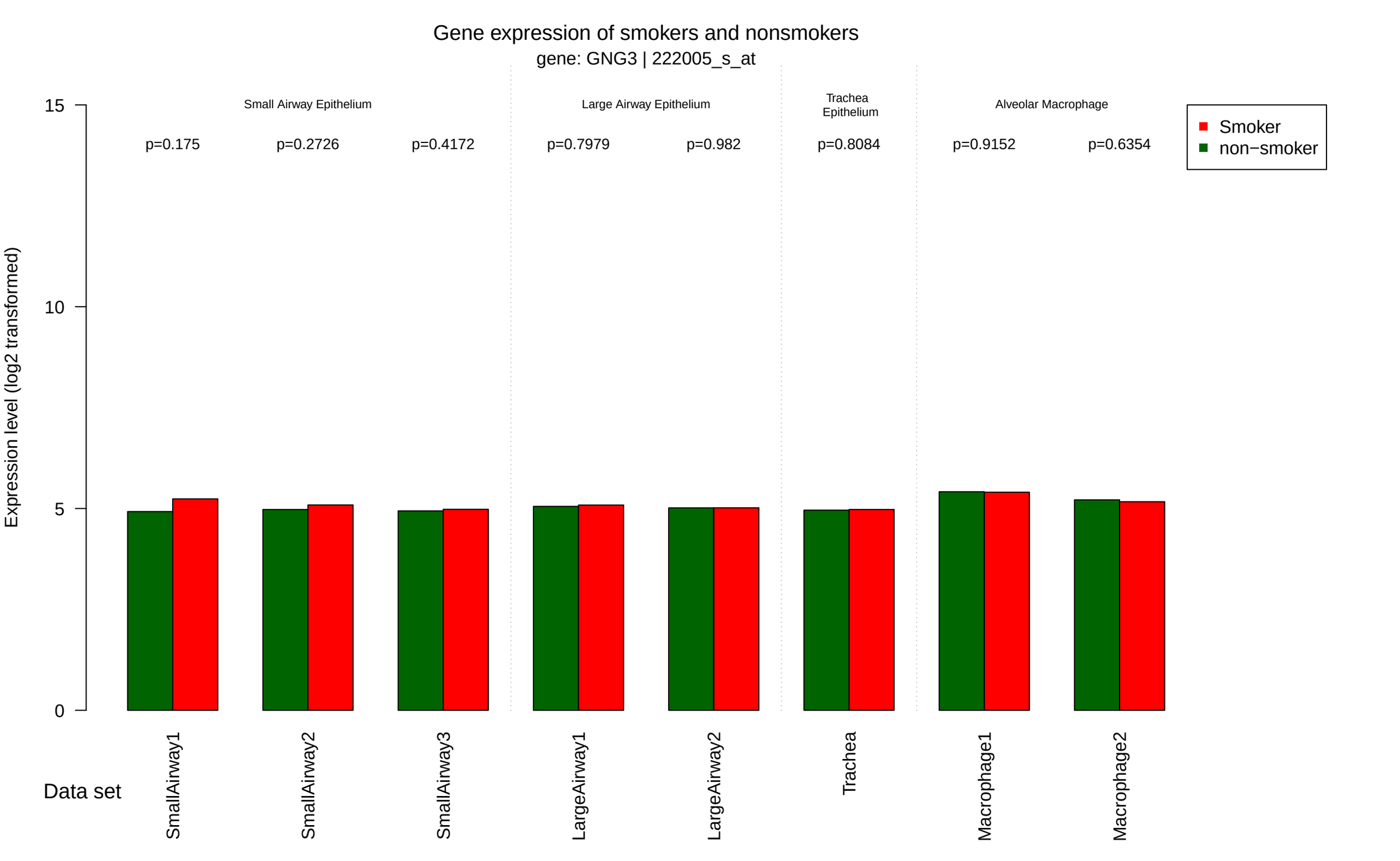
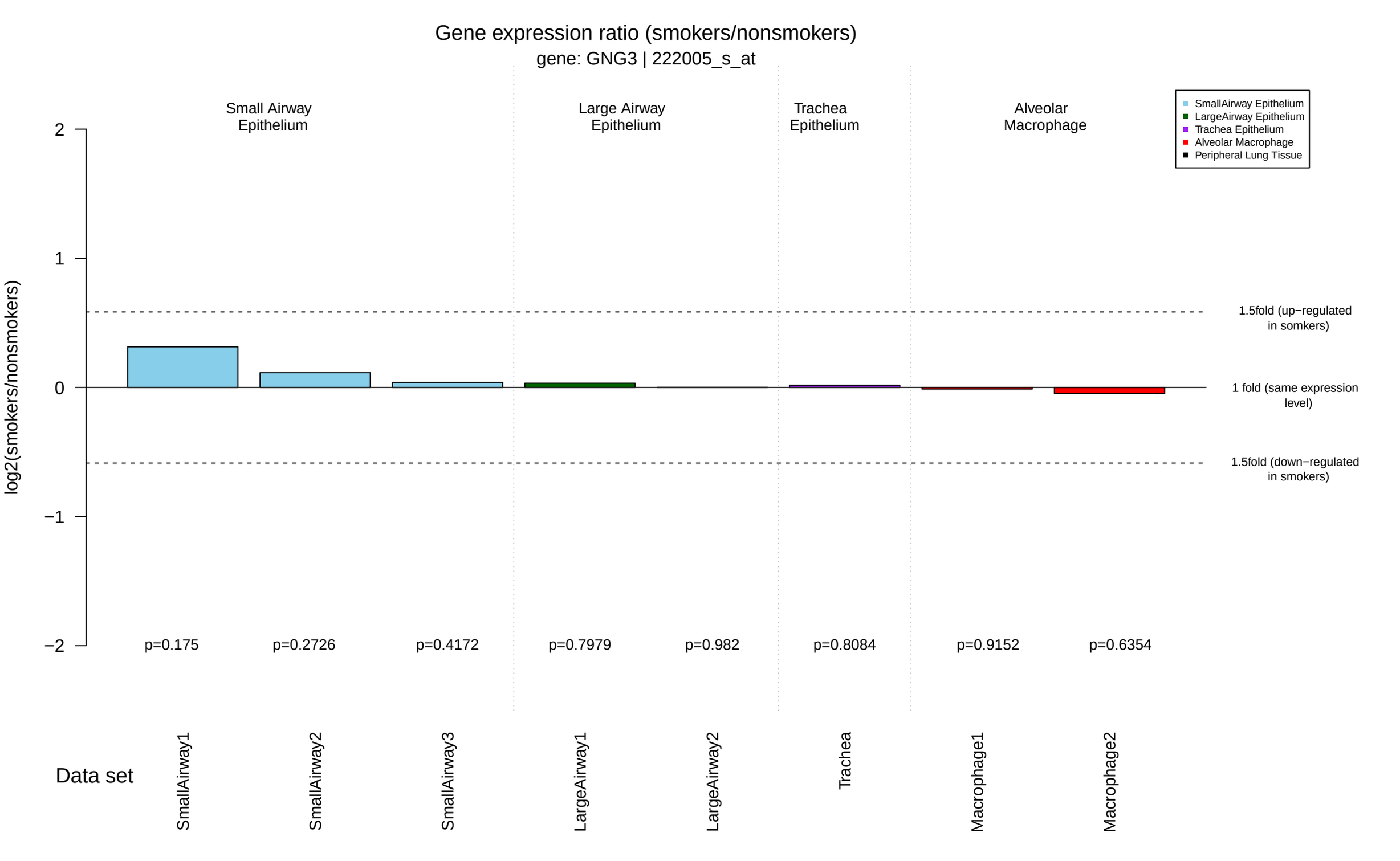
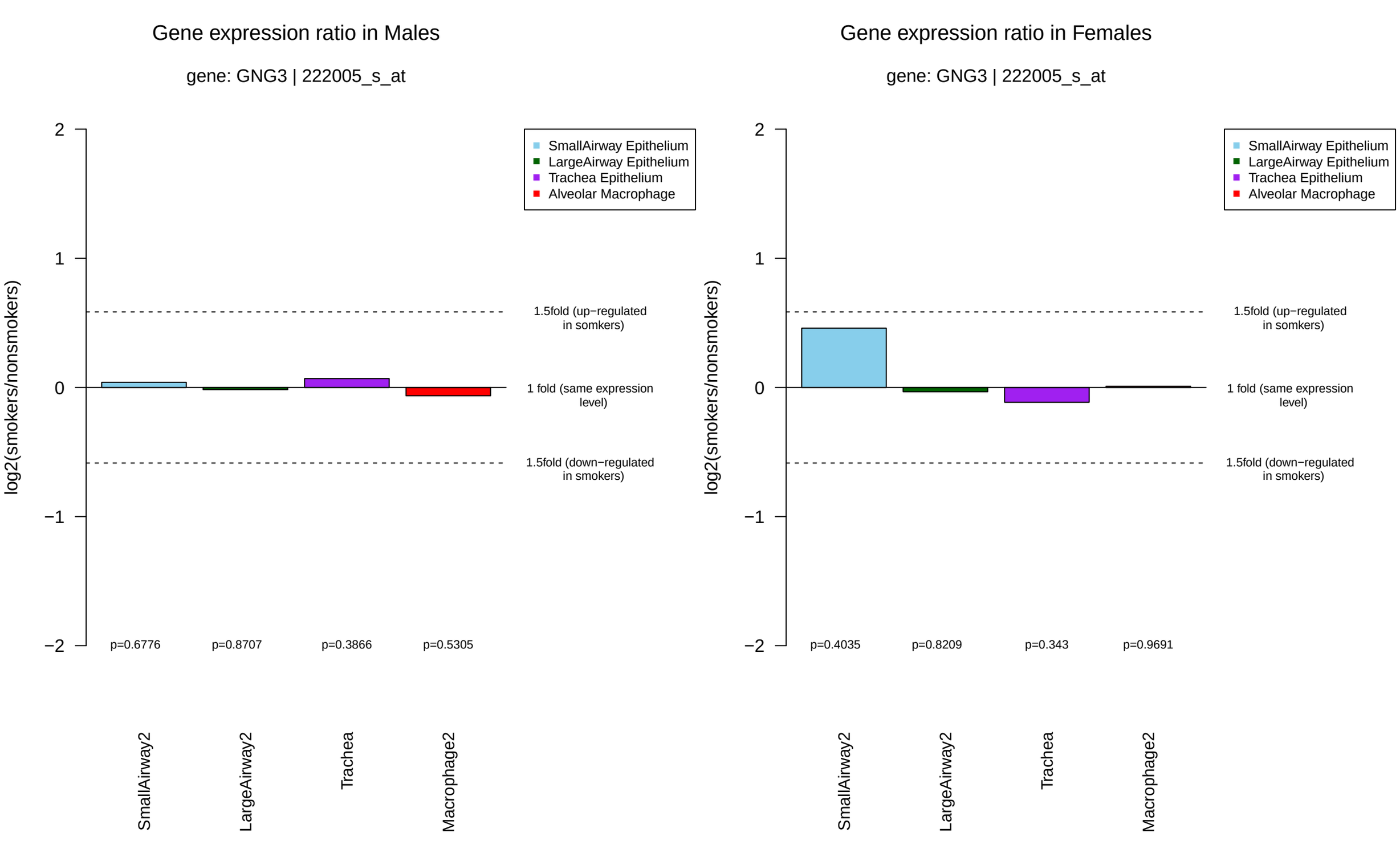
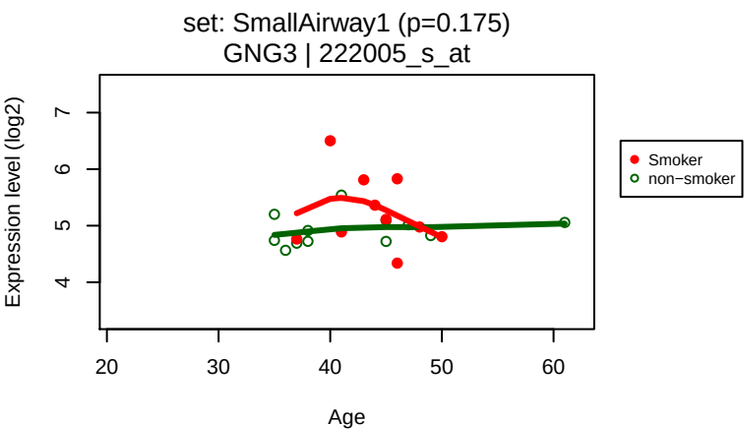
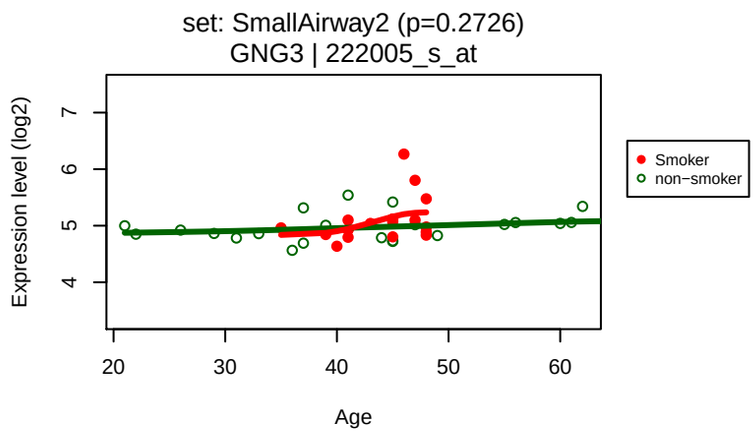
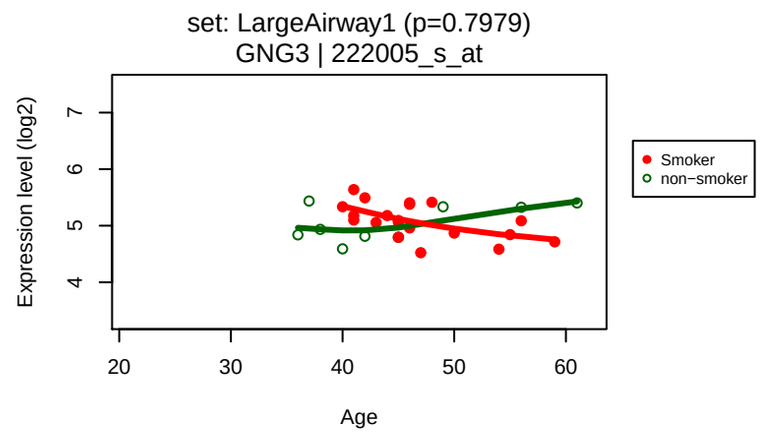

Gene expression of smokers and nonsmokers
gene: GNG3 | 222005_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.175
p=0.2726
p=0.4172
p=0.7979
p=0.982
p=0.8084
p=0.9152
p=0.6354
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

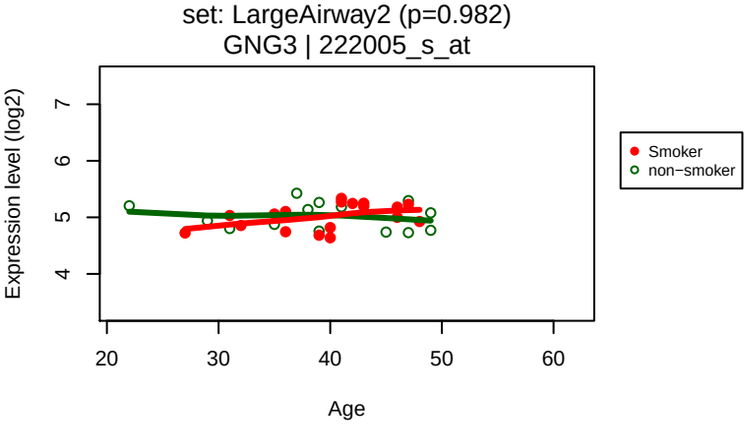

Gene expression ratio (smokers/nonsmokers)
gene: GNG3 | 222005_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.175
p=0.2726
p=0.4172
p=0.7979
p=0.982
p=0.8084
p=0.9152
p=0.6354
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

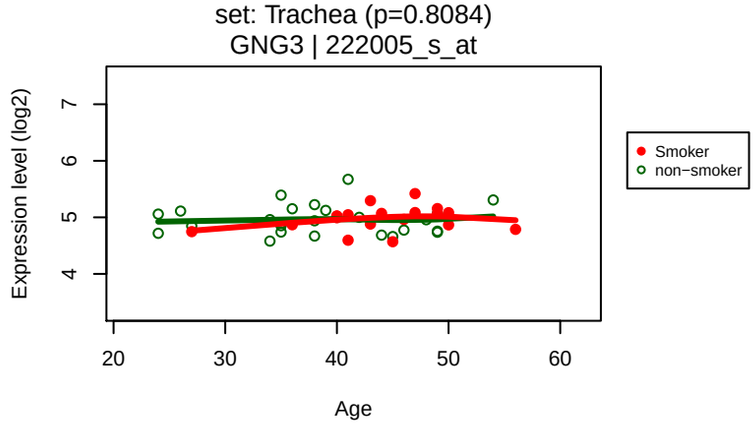

Gene expression ratio in Males
gene: GNG3 | 222005_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6776
p=0.8707
p=0.3866
p=0.5305
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNG3 | 222005_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4035
p=0.8209
p=0.343
p=0.9691
SmallAirway2
LargeAirway2
Trachea
Macrophage2

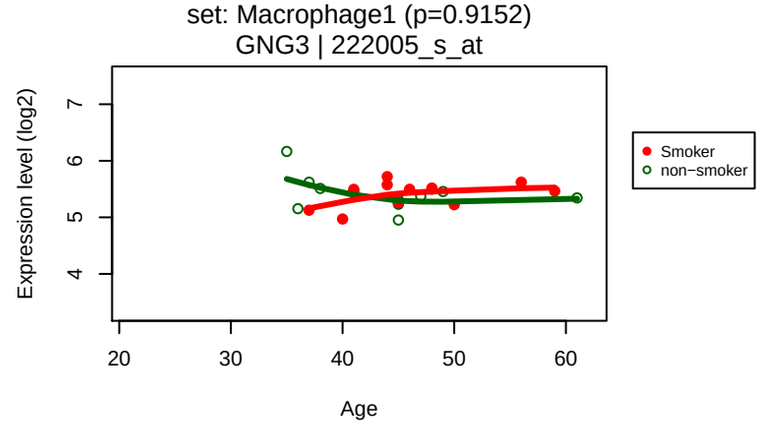

set: SmallAirway1 (p=0.175)
GNG3 | 222005_s_at
Expression level (log2)
Age
Smoker
non−smoker

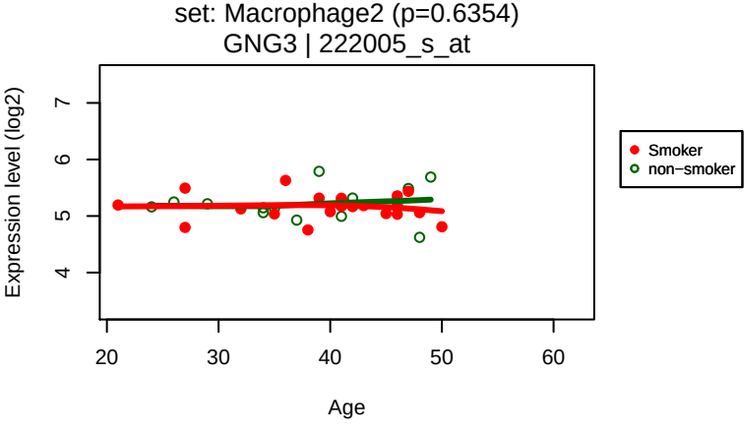

set: SmallAirway2 (p=0.2726)
GNG3 | 222005_s_at
Expression level (log2)
Age
Smoker
non−smoker

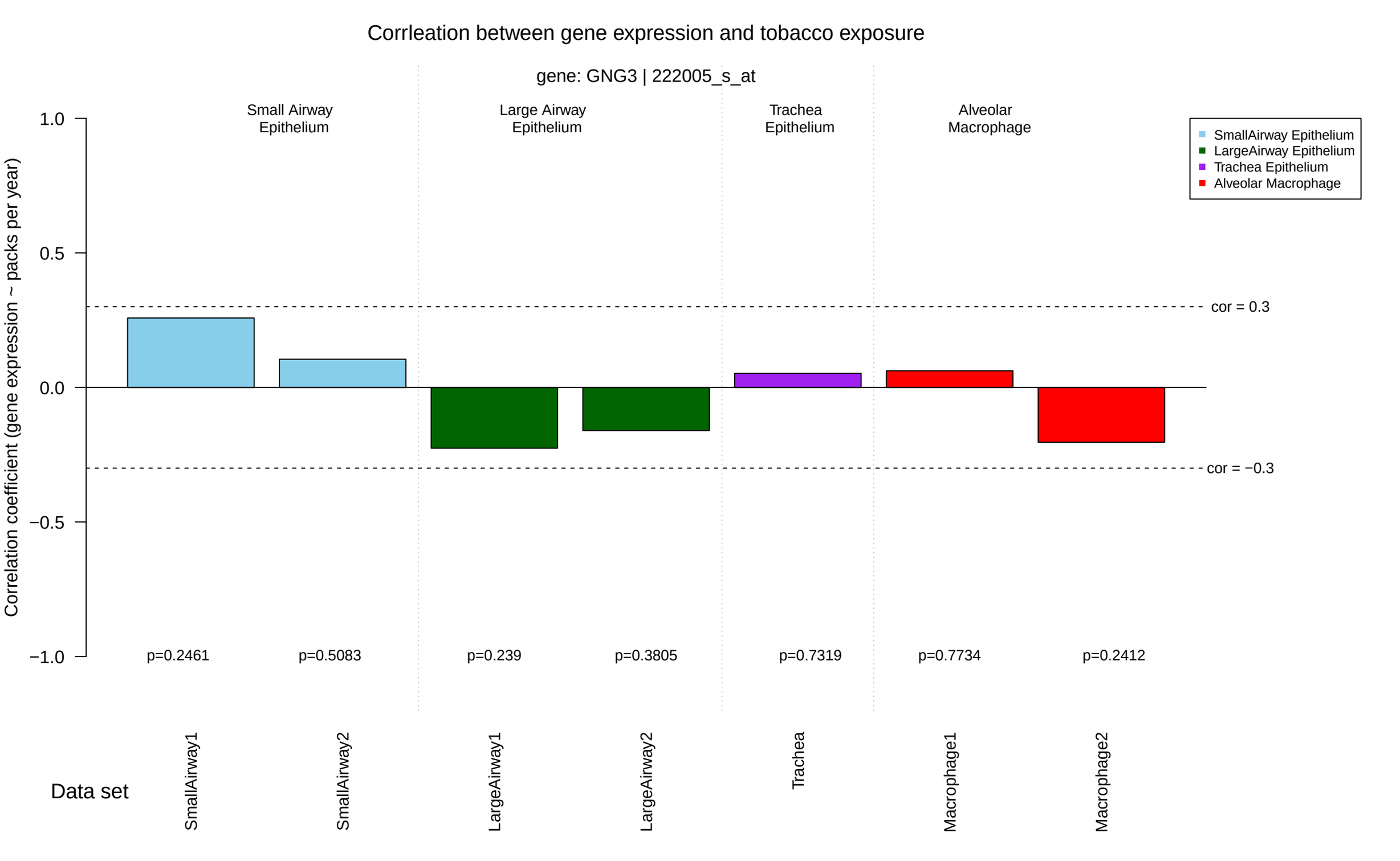

set: LargeAirway1 (p=0.7979)
GNG3 | 222005_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.982)
GNG3 | 222005_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8084)
GNG3 | 222005_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9152)
GNG3 | 222005_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6354)
GNG3 | 222005_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNG3 | 222005_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2461
p=0.5083
p=0.239
p=0.3805
p=0.7319
p=0.7734
p=0.2412
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2